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United States Patent	12403544
Kind Code	B2
Date of Patent	September 02, 2025
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Torch handle including pneumatically operated jaw

Abstract

Approaches herein provide a torch handle including a pneumatically controlled jaw operable to engage an electrode. In one approach, a system is provided for distributing gas within the torch handle, the system including a pneumatic cylinder within a main housing of the torch handle. The pneumatic cylinder is coupled to a shaft for actuating first and second members of the jaw. The system further includes a gas passageway through the main housing, the gas passageway extending to an exit orifice disposed within the second member. A flow controller, including first and second valves, is operable to direct a flow of a gas to either the pneumatic cylinder or the gas passageway. In one approach, the first member is actuated towards the second member when the gas is directed through the gas passageway, and actuated away from the second member when the gas is directed to the pneumatic cylinder.

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Appl. No.: 17/884801

Filed: August 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220379398 A1	Dec. 01, 2022

Related U.S. Application Data

division parent-doc US 15426598 20170207 US 11440123 child-doc US 17884801
division parent-doc WO PCT/US2015/048886 20150908 PENDING child-doc US 17884801

Publication Classification

Int. Cl.: B23K9/28 (20060101); B23K9/013 (20060101); B23K9/173 (20060101)

U.S. Cl.:

CPC B23K9/282 (20130101); B23K9/013 (20130101); B23K9/173 (20130101); B23K9/28 (20130101); B23K9/287 (20130101);

Field of Classification Search

CPC: B23K (9/282); B23K (9/013); B23K (9/173); B23K (9/28); B23K (9/287)

USPC: 219/138

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. non-provisional application Ser. No. 15/426,598 filed on Feb. 7, 2017, entitled “TORCH HANDLE INCLUDING PNEUMATICALLY OPERATED JAW,” which is a continuation-in-part of application serial no. PCT/US2015/048886 filed on Sep. 8, 2015, entitled “TORCH HANDLE INCLUDING PNEUMATICALLY OPERATED JAW.” The disclosures of the above applications are incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to a gouging torch. More particularly, the present disclosure relates to a gouging torch having a torch handle with a pneumatically operated jaw.

DISCUSSION OF RELATED ART

(2) Air carbon-arc systems and processes may be used in a wide variety of applications, such as metal fabrication and casting finishing, chemical and petroleum technology, construction, mining, general repair, and maintenance. With respect to metal fabrication and casting finishing, air carbon-arc metal removing systems and processes may be used to gouge, groove, cut, or flush metal from a surface.

(3) An air carbon-arc cutting and gouging torch may operate by positioning an electrode relative to a workpiece so that an electric arc can be struck and maintained between the electrode and the workpiece. As metal melts under the influence of the arc, a stream of high-pressure air directed along the electrode to the arc causes the molten metal to be forcibly removed from the influence of the arc, thus leaving a groove or gouge in the surface of the workpiece. The process can also be used to cut completely through the workpiece in the area traversed by the electrode and the arc.

(4) The electrodes of the gouging torch exhibit wear, however, and must be periodically replaced. Current art approaches use manual gouging torch levers to open and close a jaw of the torch for carbon electrode insertion and repositioning. This leads to operator fatigue, especially over a large number of cycles, as the hand gripping force necessary on the gouging torch lever may be significant.

SUMMARY OF THE DISCLOSURE

(5) Exemplary approaches herein provide a torch handle including a pneumatically controlled jaw operable to engage an electrode. In one approach, a system is provided for distributing gas within the torch handle, the system including a pneumatic cylinder disposed within a main housing of the torch handle. The pneumatic cylinder may include a piston coupled to a shaft for actuating a first member of a jaw relative to a second member of the jaw. The system further includes a gas passageway through the main housing, the gas passageway including an exit orifice disposed within the second member of the jaw. A flow controller is operable to control a flow of gas and direct it to either the pneumatic cylinder or the gas passageway. In one approach, the first member is actuated towards the second member when the flow of gas is directed through the gas passageway, and actuated away from the second member when the flow of gas is directed to the pneumatic cylinder. By providing a pneumatically assisted jaw, the hand gripping force typically required for operating the gouging torch lever may be greatly reduced.

(6) An exemplary torch handle in accordance with the present disclosure includes a main housing, and a jaw coupled to the main housing, wherein the jaw includes a first member and a second member operable to engage an electrode. The torch handle further includes a pneumatic cylinder disposed within the main housing, wherein the pneumatic cylinder is coupled to an arm for actuating the first member relative to the second member. The torch handle further includes a gas passageway extending through the main housing to an exit orifice disposed within the second member, and a valve assembly including a first valve operable to deliver a gas to the pneumatic cylinder and a second valve operable to deliver the gas to the gas passageway, wherein the first and second valves are oriented substantially perpendicular to one another.

(7) An exemplary system for distributing gas within a torch handle in accordance with the present disclosure includes a pneumatic cylinder disposed within a main housing, the pneumatic cylinder including a shaft for actuating a first member of a jaw relative to a second member of the jaw. The system further includes a gas passageway through the main housing, the gas passageway including an exit orifice disposed within the second member of the jaw. The system further includes a valve assembly including a first valve operable to deliver a gas to the pneumatic cylinder and a second valve operable to deliver the gas to the gas passageway, wherein the first and second valves are oriented substantially perpendicular to one another.

(8) An exemplary method for operating a torch handle in accordance with the present disclosure includes receiving a flow of a gas at a flow controller of a torch handle, the flow controller including a valve assembly having first and second valves oriented substantially perpendicular to one another. The method further includes directing the gas to either of the following: a pneumatic cylinder disposed within a main housing of the torch handle via the second valve, wherein the pneumatic cylinder includes a shaft for actuating a first member of a jaw, and a gas passageway through the main housing via the first valve, wherein the gas passageway includes an exit orifice disposed within a second member of the jaw. The method further includes actuating the first member of the jaw relative to the second member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate exemplary approaches of the disclosed torch handle

including a pneumatically operated jaw so far devised for the practical application of the principles thereof, and in which:

- (2) FIG. 1 is an isometric view of a torch handle according to an exemplary approach;
- (3) FIG. 2 is an isometric partial cutaway view of the torch handle of FIG. 1 according to an exemplary approach;
- (4) FIG. 3A is a side cutaway view of the torch handle of FIG. 1 according to an exemplary approach;
- (5) FIG. 3B is a close-up side cutaway view of the torch handle of FIG. 1 according to an exemplary approach.
- (6) FIG. 4 is an isometric semi-transparent view of a system for distributing gas within the torch handle of FIG. 1 according to an exemplary approach;
- (7) FIG. 5 is an isometric view of a valve of a flow controller within the torch handle of FIG. 1 according to exemplary approaches.
- (8) FIG. 6-A is an isometric partial cutaway view of the torch handle of FIG. 1 according to an exemplary approach;
- (9) FIG. 6-B is a side cutaway view of the torch handle of FIG. 1 according to an exemplary approach;
- (10) FIG. 7-A is a semitransparent, isometric partial cutaway view of the torch handle of FIG. 1 according to an exemplary approach;
- (11) FIG. 7-B is a semitransparent, side cutaway view of the torch handle of FIG. 1 according to an exemplary approach; and
- (12) FIG. 8 is a flow chart illustrating an exemplary method of operating a gouging torch having a torch handle with a pneumatically operated jaw.

DESCRIPTION OF EMBODIMENTS

(13) The present disclosure will now proceed with reference to the accompanying drawings, in which various approaches are shown. It will be appreciated, however, that the disclosed torch handle may be embodied in many different forms and should not be construed as limited to the approaches set forth herein. Rather, these approaches are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. In the drawings, like numbers refer to like elements throughout.

(14) As used herein, an element or operation recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or operations, unless such exclusion is explicitly recited. Furthermore, references to “one approach” of the present disclosure are not intended to be interpreted as excluding the existence of additional approaches that also incorporate the recited features.

(15) Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “central,” “above,” “upper,” and the like, may be used herein for ease of describing one element's relationship to another element(s) as illustrated in the figures. It will be understood that the spatially relative terms may encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures.

(16) Referring now to FIG. 1, a torch handle **10** of a gouging torch according to exemplary embodiments will be described in greater detail. As shown, the torch handle **10** includes a main housing **14**, which may include one or more pieces assembled together and suitably shaped to be held in a hand of an operator. The torch handle **10** further includes a jaw **22** coupled at a distal end **18** of the main housing **14**. Although not shown, the torch handle **10** includes a torch body disposed within the main housing **14** at a proximal end **20** thereof. In one embodiment, the torch handle **10** and the jaw are made of a heat resistant material such as a glass-filled phenolic.

(17) In an exemplary embodiment, the jaw **22** includes a first member **24** disposed opposite a second member **26**. Together the first member **24** and the second member **26** may operate to engage an electrode **30** therebetween. In exemplary embodiments, the jaw **22** may be user-controlled via a

switch **34** that is partially recessed within an opening **38** of the main housing **14**. More specifically, the switch **34** may operate with a flow controller within the main housing **14** to provide pneumatically assisted user-control of the jaw **22**, as will be further described below.

(18) In exemplary embodiments, the first member **24** includes a protrusion **32** configured to make contact with the electrode **30** when the jaw **22** is in a closed position. The second member **26** includes a nozzle **36** having a groove formed therein to receive the electrode **30**. The nozzle **36** may be free to move with the movement of electrode **30**. Nozzle **36** is provided with a plurality of orifices **31** from which flows a high velocity gas stream which performs the cutting or gouging action. The pressure of gas required to perform the work varies, but usually this torch operates from a gas source having a pressure of from about 80 to about 100 psi.

(19) Referring now to FIGS. 2-3B, the torch handle **10** according to exemplary embodiments will be described in greater detail. As shown, the torch handle **10** includes a pneumatic cylinder **40** disposed within the main housing **14**. The pneumatic cylinder **40** includes a shaft **42** coupled to an arm **44** at a first end, and to a piston **46** disposed within a chamber **48** at a second end. In exemplary embodiments, the piston **46** is configured to slide within the chamber **48** in response to a flow of gas through a pneumatic cylinder inlet **50**. The piston **46** may include one or more O-rings **53** extending around a circumference thereof to create a seal between the piston **46** and the walls of the chamber **48**.

(20) In various embodiments, the pneumatic cylinder **40** may differ in appearance, size and function depending upon the type of pneumatic cylinder employed. For example, in one embodiment, a single-acting cylinder (SAC) may use the pressure imparted by compressed air to create a driving force in one direction, while a spring is used to return the piston **46** to its original position. In another embodiment, a double-acting cylinder may use the force of air to move the piston **46** in both directions. The cylinder may have two ports to allow air into the cylinder, one for outstroke and one for instroke. In yet another embodiment, a multi-stage, telescoping cylinder, which may be either single or double-acting, incorporates a piston rod nested within a series of hollow stages of increasing diameter. Upon actuation, the piston rod and each succeeding stage “telescopes” out as a segmented piston. This allows for a longer stroke than would be achieved with a single-stage cylinder of the same collapsed length.

(21) As further shown, the arm **44** includes one or more slots **52** for slidably receiving a radial pin **54** of the shaft **42**. In exemplary embodiments, the arm **44** includes a first section **60** disposed within the main housing **14**, and a second section **62** coupled to the first member **24** of the jaw **22**. During operation, the arm **44** may pivot about a fulcrum **58** in response to movement of the radial pin **54** within the slot **52**. For example, when the shaft **42** is urged by the piston **46** towards the distal end **18** of the main housing **14**, the radial pin **54** is positioned at a top end of the slot **52**, which causes the first section **60** of the arm **44** to move downward in a direction generally transverse to the linear movement of the piston **46**, and the second section **62** of the arm **44** to move upwards away from the electrode **30**. Inversely, when the shaft **42** is retracted towards the proximal end **20** of the main housing **14**, the radial pin **54** is positioned at a bottom end of the slot **52**, which causes the first section **60** of the arm to move upwards in the direction generally transverse to the movement of the piston **46**, and the second section **62** of the arm **44** to move downward towards the electrode **30**.

(22) In one embodiment, the torch handle **10** further includes a spring **64** coupled to the arm **44** of the first member **24**. As shown, the spring **64** is oriented generally perpendicular to the movement of the piston **46**, which causes the first section **60** of the arm to be biased in the direction generally transverse to the movement of the piston **46**, and the second section **62** of the arm **44** to move towards the electrode **30**. During use, the spring **64** applies a light force to the first member **24** sufficient enough to secure the electrode **30** between the nozzle **36** and the protrusion **32** when the jaw **22** is open.

(23) Referring now to FIGS. 3A-4, a system **66** for pneumatically operating the jaw **22** by

controlling distribution of a gas within the torch handle **10** according to exemplary embodiments will be described in greater detail. As shown, the system **66** includes the pneumatic cylinder **40** and a gas passageway **68** through the main housing **14**, the gas passageway **68** including one or more exit orifices **31** (FIG. **1**) disposed within the nozzle **36** of the second member **26** of the jaw **22**. During use, gas under pressure released from exit orifices **31** passes alongside the electrode **30** to remove molten metal from the work surface in the area of the electric arc. In one embodiment, each orifice **31** is oriented parallel to the electrode **30** to maintain sufficient force and to effect the stream of gas acting on the work surface.

(24) As shown, a supply of gas **72** for operating the torch handle **10** is directed through the main housing **14** and received at a gas inlet **74**. The supply of gas **72** may be compressed air, which is delivered to a flow controller **78** for further distribution within the torch handle **10**. In an exemplary embodiment, the flow controller **78** is a valve assembly **79** including a first valve **82** and a second valve **84**, wherein the first valve **82** is disposed within a valve chamber **81** and configured to direct a gas flow **73** to the pneumatic cylinder inlet **50** via the first valve **82**, or to the gas passageway **68** via the second valve **84** in response to a desired torch operational mode (e.g., “gouging” or “release”) selectable via the switch **34**. In one embodiment, the torch handle **10** may also include an “OFF” mode that shuts off the supply of gas **72**.

(25) In one embodiment, in the case that the gouging mode is selected, the gas flow **73** may be directed through the gas passageway **68**, which decreases or maintains a relatively lower pressure within the chamber **48**, and which slides the piston **46** towards the proximal end **20** of the torch handle **10**, thus actuating the first member **24** of the jaw **22** towards the second member **26**. In the case that the release mode is selected, the flow controller **78** may divert the gas flow **73** to the pneumatic cylinder **40**, which directs the gas flow **73** to the chamber **48**. A resultant pressure increase within the chamber **48** actuates the piston **46** towards the distal end **18** of the torch handle **10**, which causes the first member **24** to move away from the second member **26**.

(26) In some embodiments, the first valve **82** and the second valve **84** of the valve assembly **79** are oriented perpendicular, or generally perpendicular, to one another. That is, a lengthwise central longitudinal axis **57** of the first valve **82** may be perpendicular to a lengthwise central longitudinal axis **59** of the second valve **84**. As shown, the lengthwise central longitudinal axis **57** is generally perpendicular to a lengthwise axis (not shown) of the torch handle **10**. By orienting one of the first valve **82** on its side relative to the second valve **84**, the first valve **82** may be made larger, which improves performance of the torch handle **10** by increasing gas flow through the gas passageway **68**. A larger valve oriented parallel to the lengthwise central longitudinal axis **59** of the second valve **84** may be impractical due to constraints on operator preferred handle size. In some embodiments, the first valve **82** has a length and a circumference that is greater than a length and a circumference of the second valve **84**. The valve assembly **79** may include a greater or fewer number of valves. For example, multiple valves may be substituted for the first valve **82** in alternative designs.

(27) In some embodiments, the switch **34** may be a rocker switch including a main body **33** and a lever arm **35** extending from the main body **33**, the lever arm **35** for controlling/operating each of the first and second valves **82**, **84** of the valve assembly **79**. As best shown in FIG. **3B**, the main body **33** may extend outside of the main housing **14** of the torch handle **10** for access by a user, while the lever arm **35** extends into the torch handle, for example, perpendicular to the main body **33**. The main body **33** may include a first end **37** and a second end **39**, wherein the first or second ends **37**, **39** may be depressed to cause the main body **33** to pivot about a pin **41**, controlling the gas flow **73**. In some embodiments, the pin **41** may be suspended or supported by a switch support **43**.

(28) The switch **34** may further include a cantilever arm **45** coupled to the main housing and in direct physical contact with a stem **100-A** of second valve **84**. The cantilever arm **45** is further coupled to a first side of the lever arm **35**. A second side of the lever arm **35** is coupled to a stem **100-B** of the first valve **82**. During use, when the second end **39** of the switch **34** is depressed into a

first cavity **47**, the lever arm **35** rotates towards the stem **100-B** of the first valve **82** and away from the cantilever arm **45**. The lever arm **35** may bias the stem **100-B** towards the proximal end **20** and into the first valve **82**, as will be described in greater detail below. Inversely, when the first end **37** of the switch **34** is depressed into a second cavity **49**, the lever arm **35** rotates towards the cantilever arm **45**. A force applied to the free end of cantilever arm **45** causes the cantilever arm **45** to rotate and bias against the stem **100-A** of the second valve **84**, causing the stem **100-A** to depress within the second valve **84**, as will be described in greater detail below. A stem spring **51** may surround the stem **100-A** of the second valve **84** to apply a spring force against the cantilever arm **45**.

(29) Referring now to FIG. 5, operation of the first valve **82** and the second valve **84** of the valve assembly **79** according to an exemplary embodiment will be described in greater detail. For the sake of simplicity, only the second valve **84** is shown. However, it'll be appreciated that the first valve **82** may operate with similar elements and in a similar way. In this embodiment, the second valve **84** represents a normally closed 3-way cartridge valve having a base section **86** including an inlet **87** at a first end **89** thereof, and a central section **88** separated from the base section **86** by an O-ring **90**. As shown, the central section **88** includes a conduit **94** formed therethrough. In exemplary embodiments, the central section **88** and the conduit **94** are aligned with the pneumatic cylinder inlet **50** (FIGS. 3A-B). The second valve **84** further includes an upper section **96** separated from the central section **88** by an O-ring **98**, and a stem **100-A** disposed at a second end **101** thereof. In exemplary embodiments, the stem **100-A** is configured to change a gas flow path when depressed and/or released. As shown, the stem **100-A** may include an exhaust **102** for dispelling high pressure gas from the stem **100-A**.

(30) In exemplary embodiments, as described above, the switch **34** causes the cantilever arm **45** to engage the stem **100-A** so as to cause the stem **100-A** to be recessed to a desired depth within the second valve **84**. For example, in the case that the second valve **84** is a three-position valve, the switch **34** may recess the stem **100-A** a distance **D1** relative to a top surface **91** of the upper section **96**. In this first position, a passage (not shown) of the stem **100** aligns with the conduit **94** in the central section **88** of the first valve **82**, **84** to expel gas radially therefrom. In another position, the switch **34** may recess the stem **100-A** a distance **D2** relative to the top surface **91** of the upper section **96** such that the exhaust **102** is recessed within the first valve **82** proximate the upper section **96** to prevent fluid flow from the exhaust **102** and to cause the passage of the stem to be misaligned with the conduit **94**. In this position, gas within the second valve **84** is prevented from being released via the exhaust **102** and/or the conduit **94**.

(31) Referring now to FIGS. 6A-B, further operation of the valve assembly **79** according to exemplary embodiments will be described in greater detail. As shown, the valve assembly **79** includes the second valve **84** configured to receive the supply of gas **72** through the gas inlet **74** and deliver it to the chamber **48** via the pneumatic cylinder inlet **50** for the purpose of changing the relative positions of the first and second members of the jaw **22**, as described above. In this embodiment, the supply of gas **72** passes through the gas inlet **74** and surrounds the base section **86-A** of the second valve **84**. To prevent gas from moving along an exterior surface of the second valve **82** toward the central section **88-A**, the O-ring **90-A** is provided. As shown, the O-ring **90-A** forms a seal with an interior wall of a valve chamber **83**. The gas is thus directed to the inlet **87** (FIG. 5) where, depending on the position of the stem **100-A**, it may be directed through the conduit **94-A**, and then through the pneumatic cylinder inlet **50**. In the case that the gas is prevented from being expelled through the conduit **94-A**, the gas is re-directed along the exterior of the base section **86-A** towards the first valve **82**.

(32) During operation, to direct the gas to the chamber **48**, the stem **100-A** is depressed relative to the top surface **91-A** of the first valve **82**, and an opening (not shown) along the stem **100-A** aligns with the conduit **94-A**. The gas traverses an interior conduit (not shown) within the stem **100-A** where it is expelled radially through the conduit **94-A**. The gas surrounds the central section **88-A**

of the second valve **84**, where it is confined by the lower O-ring **90-A**, the upper O-ring **98-A**, and the valve chamber **83**. The gas may then enter the pneumatic cylinder inlet **50**, where it is delivered to the chamber **48**. The resultant pressure increase within the chamber **48** caused by the flow of gas actuates the piston **46** in a linear direction away from the second valve **84**, which causes the first member of the jaw to move away from the second member of the jaw, for example, to allow for repositioning or replacement of the electrode between gouging operations.

(33) Referring now to FIGS. **7A-B**, operation of the valve assembly **79** will be described in greater detail. As shown, the valve assembly **79** includes the first valve **82** configured to receive the supply of gas **72** through the gas inlet **74** and deliver it to the gas passageway **68** for the purpose of providing gas to the nozzle **36** (FIGS. **1-3**), as described above. More specifically, gas is directed through the gas inlet **74** towards the valve chamber **81** where it surrounds the base section **86-B** of the first valve **82**. The O-ring **90-B** forms a seal with a wall of the valve chamber **81** that extends around the circumference of the first valve **82** to prevent the supply of gas from flowing along the exterior of the first valve **82** toward the central section **88-B**. The gas may thus enter the inlet **87** (FIG. **5**) where, depending on the position of the stem **100-B**, it may be directed through the conduit **94-B**. In the case that the gas is prevented from being expelled through the conduit **94-B**, the gas is re-directed along the exterior of the base section **86-B** towards the second valve **84**.

(34) To direct the gas to the gas passageway **68**, the stem **100-B** is depressed relative to the top surface **91-B** of the first valve **82**, and an opening (not shown) along the stem **100-B** aligns with the conduit **94-B**. The gas traverses an interior conduit (not shown) of the stem **100-B** where it is expelled radially through the conduit **94-B**. The gas thus surrounds the central section **88-B** of the first valve **82** where it is confined by the lower O-ring **90-B**, the upper O-ring **98-B**, and the valve chamber **81**. The gas may then enter an interior conduit **108** of the gas passageway **68** where it is delivered to the nozzle **36** (FIGS. **1-3**). In exemplary embodiments, pressure within the gas passageway **68** increases due to the flow of the gas, which causes a resultant pressure decrease within the chamber **48** of the pneumatic cylinder **40**. This decrease in pressure allows the piston **46** to actuate within the chamber **48** in a direction towards the second valve **84**, which causes the first member of the jaw to move towards the second member of the jaw.

(35) Referring now to FIG. **8**, a method **200** for operating a torch handle according to exemplary embodiments will be described in greater detail. Method **200** includes receiving a flow of a gas at a flow controller of the torch handle, as shown at block **201**. In some embodiments, the gas may be compressed air, which is provided to the flow controller via a gas inlet of the torch handle. In some embodiments, the flow controller is a valve assembly including first and second valves. In some embodiments, the first and second valves are oriented perpendicular to one another.

(36) The method **200** further includes directing a flow of the gas to either a pneumatic cylinder disposed within a main housing via a second cylinder, or a gas passageway via a first cylinder, as shown at block **203**. In some embodiments, the pneumatic cylinder includes a piston have a shaft configured to actuate a first member of a jaw. In some embodiments, the gas passageway extends through the main housing to an exit orifice disposed within a second member of the jaw. In some embodiments, a switch is coupled to the flow controller for allowing user-operation of the flow controller. In some embodiments, the shaft of the piston is coupled to an arm, which is partially disposed within the first member and configured to actuate the first member.

(37) The method **200** further includes actuating the first member of the jaw relative to the second member of the jaw in response to the flow of the gas, as shown in block **205**. In some embodiments, the first member is actuated towards the second member when the flow of gas is directed through the gas passageway, and actuated away from the second member when the flow of gas is directed to the pneumatic cylinder.

(38) As will be appreciated, it is advantageous to replace a manual lever with a pneumatic cylinder and valve assembly, as described herein, to aid the opening and closing of the torch jaw for carbon electrode insertion and repositioning. Furthermore, as compressed air flows to the nozzle during a

gouging process, it is advantageous to include a valve assembly, as described herein, for diverting the compressed air to the pneumatic cylinder to open the jaw. It will be further appreciated that use of the valve assembly and pneumatic cylinder in place of a manual level advantageously allows for a reduction in handle diameter, which improves user ergonomics and reduces user fatigue.

(39) While the present disclosure has been described with reference to certain approaches, numerous modifications, alterations and changes to the described approaches are possible without departing from the sphere and scope of the present disclosure, as defined in the appended claims. Accordingly, it is intended that the present disclosure not be limited to the described approaches, but that it has the full scope defined by the language of the following claims, and equivalents thereof. While the disclosure has been described with reference to certain approaches, numerous modifications, alterations and changes to the described approaches are possible without departing from the spirit and scope of the disclosure, as defined in the appended claims. Accordingly, it is intended that the present disclosure not be limited to the described approaches, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

Claims

1. A torch handle, comprising: a main housing; a jaw coupled to the main housing, the jaw including a first member and a second member operable to engage an electrode; a pneumatic cylinder disposed within the main housing, the pneumatic cylinder coupled to an arm for actuating the first member relative to the second member; a gas passageway extending through the main housing to an exit orifice disposed within the second member; and a valve assembly including a first valve operable to deliver a gas to the pneumatic cylinder and a second valve operable to deliver the gas to the gas passageway, wherein the first and second valves are oriented perpendicular to one another.
2. The torch handle of claim 1, further comprising a gas inlet for delivering a supply of the gas to the valve assembly.
3. The torch handle of claim 1, wherein the first valve has a length and a circumference greater than a length and a circumference of the second valve.
4. The torch handle of claim 1, further comprising a switch coupled to the valve assembly for providing user-operation of the jaw, the switch comprising: a main body; and a lever arm extending from the main body, the lever arm in direct physical contact with at least one of the first and second valves.
5. The torch handle of claim 4, wherein the main body extends partially outside of the main housing for access by a user, and wherein while the lever arm extends into an interior of the torch handle.
6. The torch handle of claim 4, the switch further comprising a cantilever arm coupled to the main housing, wherein the cantilever arm is in direct physical contact with a stem of the second valve, and wherein rotation of the cantilever arm actuates the stem of the second valve.
7. The torch handle of claim 1, the pneumatic cylinder comprising: a piston within a chamber; and a pneumatic cylinder inlet for delivering the gas to the chamber.
8. The torch handle of claim 1, the arm comprising a slot for slidably receiving a shaft of the pneumatic cylinder.
9. The torch handle of claim 1, further comprising a spring coupled to the arm of the first member.
10. The torch handle of claim 8, wherein the arm includes a first section disposed within the main housing and a second section coupled to the first member, the slot of the arm being disposed on the first section.
11. The torch handle of claim 10, wherein the arm is configured to pivot about a fulcrum disposed between the first section and the second section.
12. The torch handle of claim 1, wherein the second member includes a nozzle comprising a

plurality of orifices.

13. The torch handle of claim 12, wherein the nozzle further comprises a groove configured to receive the electrode.

14. The torch handle of claim 13, wherein the nozzle is configured to move with movement of the electrode.

15. The torch handle of claim 13, wherein the first member includes a protrusion configured to contact the electrode received by the groove of the nozzle when the first member is in a closed position.
